# Evaluate Binary Log Loss

## Description

Calculates mean binary cross-entropy after clipping probabilities away from zero and one.

## Usage

```
evaluate_binary_log_loss(
  observed = NULL,
  predicted_probability = NULL,
  epsilon = 1e-06
)
```

## Arguments

| | |
|---|---|
| `observed` | Numeric binary observation vector. |
| `predicted_probability` | Numeric predicted-probability vector aligned with 'observed'. |
| `epsilon` | Single numeric clipping tolerance strictly between zero and '0.5'. Defaults to '1e-6'. |

## Value

One-row tibble containing 'log_loss', 'metric_status', observation counts, prevalence, and the clipping tolerance. Valid one-class inputs retain a finite log loss because this metric does not require both classes.

## Examples

```
evaluate_binary_log_loss(
  observed = c(0, 1),
  predicted_probability = c(0.25, 0.75)
)
```